DO NOW — RECALL

Lesson 054 · Hash Tables and Dictionary<TKey,TValue>

Name: _____ Class: _____ Date: _____

1. Which data structure uses FIFO in BFS?

2. What does a dictionary map?

3. What is a collision in hashing?

4. A hash table has slots 0–6, all empty. Insert keys 9, 16 and 23 in that order using h(k)=k MOD 7 and linear probing to the next slot. Determine the final slot occupied by 23.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A queue.

2. Model answer

Keys to associated values.

3. Model answer

Different keys map to same hash location.

4. Model answer

Slot 4. All three hash to 2; 9 occupies 2, 16 occupies 3, and 23 probes to 4.

By the end, you can…

• A hash function maps keys to hash values that guide placement/lookups. A collision occurs if distinct keys map to the same location/bucket.

• A .NET Dictionary<TKey,TValue> handles collision management internally; do not equate its actual buckets with simplistic modulo examples.

• To read a missing key safely, use TryGetValue. Hash-table operations are often expected O(1) but poor collisions can degrade performance.

Vocabulary — know these terms

hash function — Maps keys to hash values.

collision — Different keys share a target bucket.

bucket — Hash table storage location.

load factor — Filled entries relative to capacity.

NEW LEARNING

Read, annotate and explain

Discover

A hash function maps keys to hash values that guide placement/lookups. A collision occurs if distinct keys map to the same location/bucket.

Trace

A .NET Dictionary<TKey,TValue> handles collision management internally; do not equate its actual buckets with simplistic modulo examples.

Evaluate

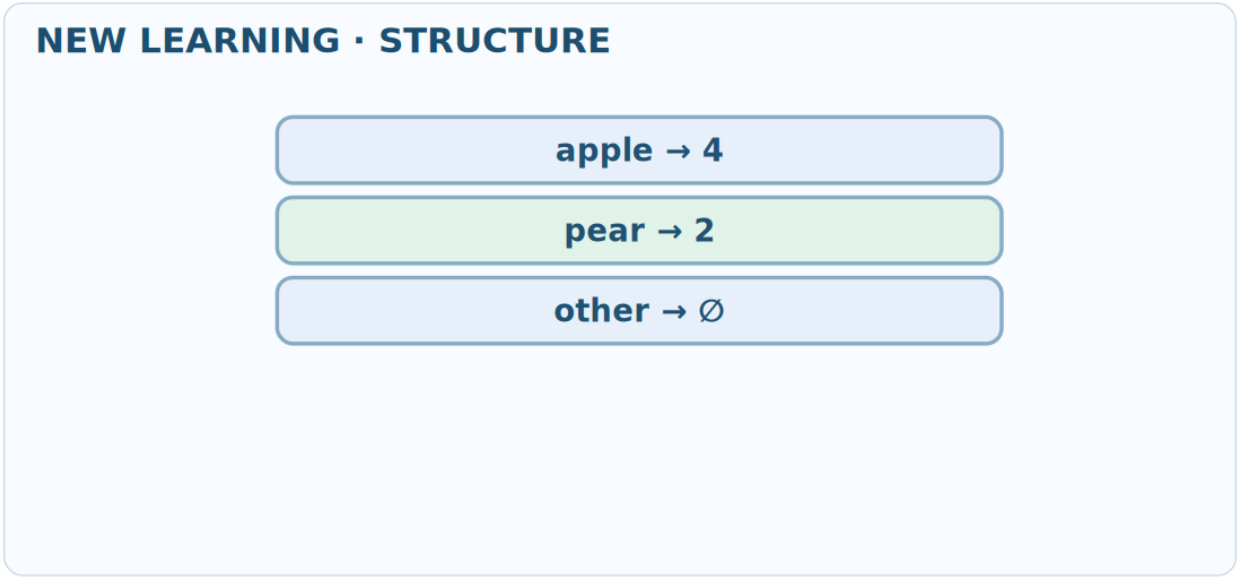
To read a missing key safely, use TryGetValue. Hash-table operations are often expected O(1) but poor collisions can degrade performance.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



NEW LEARNING · STRUCTURE: apple → 4, pear → 2, other → ∅

I DO — WORKED EXAMPLE

Follow the worked method

Apply a toy hash h(k)=k mod 5 to keys 7 and 12. What happens?

Explain the trace

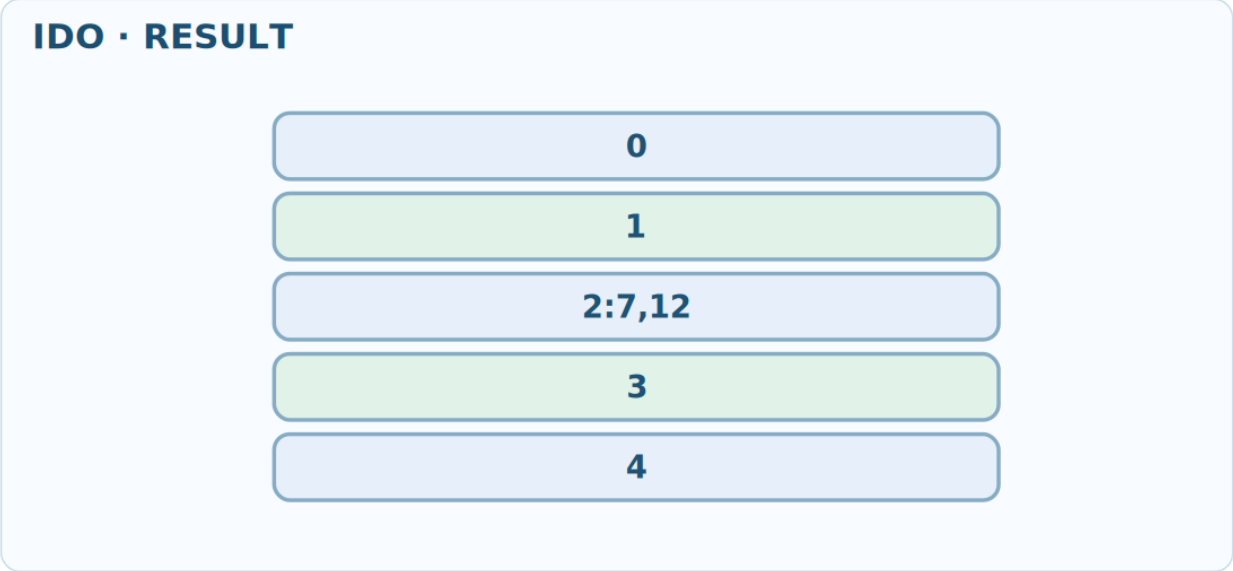
Both keys collide in bucket 2; a table must resolve collision by chaining or probing.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

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Both keys collide in bucket 2; a table must resolve collision by chaining or probing.



IDO · RESULT: 0, 1, 2:7,12, 3, 4

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Apply the same toy hash h(k)=k mod 5 to keys 8 and 11. Give buckets.
Source code / input:
h(8)=3
h(11)=1

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

With toy h(k)=k mod5, find buckets for keys 8,13,19. [6]

Question 2

Write C# Dictionary<string,int> code storing marks Ada80 and Ben65, then safely test whether Cleo exists. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Compare chaining and linear probing for toy hash collisions, and state two performance concerns. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Hash tables can never have collisions when keys are different.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

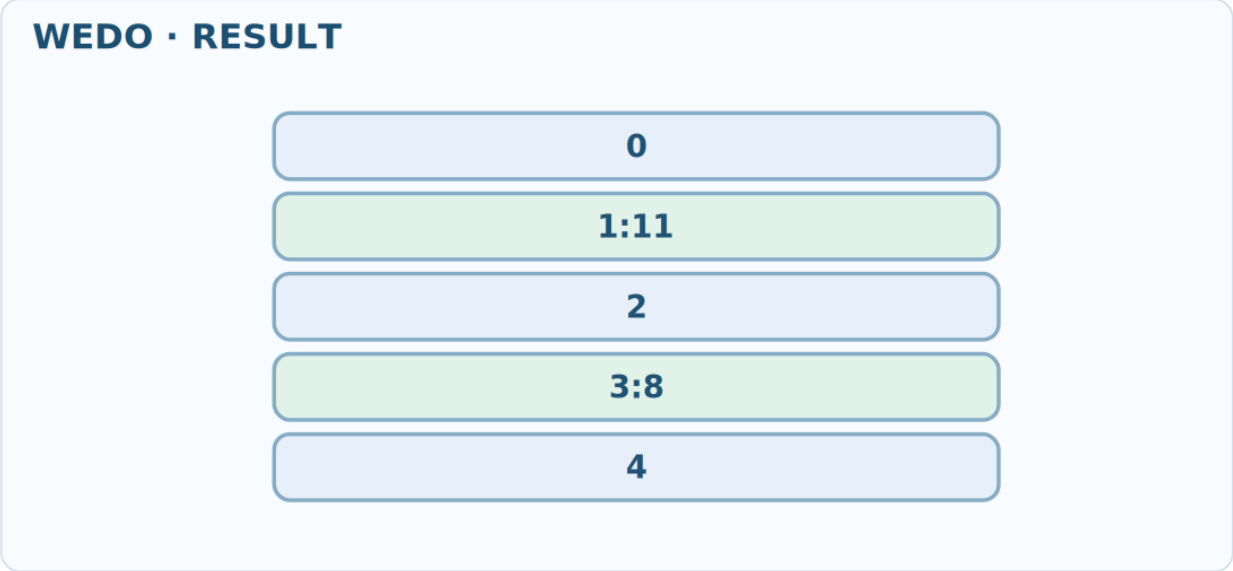
Explain the trace
No collision: key 8 maps bucket 3, key11 maps bucket1.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

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No collision: key 8 maps bucket 3, key11 maps bucket1.



WEDO · RESULT: 0, 1:11, 2, 3:8, 4